Chapter 6: Annual Worth Analysis

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Instructions:

- For every calculation, clearly state which interest factor(s) you are using in standard notation (e.g., $(A/P, i, n)$) before substituting numerical values.
- Show the formula you are using before computing the final numerical answer.
- Round final dollar answers to the nearest cent, unless otherwise stated.

Exercise 1: Capital Recovery and Annual Worth of a Clinical Lab Analyzer

A diagnostic laboratory is evaluating the purchase of a new blood analyzer. The analyzer costs \$380,000 to purchase and install, has a useful life of 5 years, and is expected to have a salvage value of \$40,000 at the end of its life. The analyzer is projected to generate additional net annual revenue of \$115,000. The lab's MARR is 9% per year.

1. Compute the Capital Recovery (CR) of the analyzer.
2. Compute the Annual Worth (AW) of the analyzer.
3. Interpret your AW result in one or two sentences: what does it tell the lab about the analyzer's economic viability, and should the lab proceed with the purchase?

Exercise 2: Annual Worth of a Cost Alternative

A regional airport is required by regulation to provide a de-icing service for aircraft during winter months. Management is evaluating a new de-icing rig that costs \$540,000 to purchase, has a useful life of 8 years with no salvage value, and is expected to cost \$65,000 per year to operate and maintain. The airport's MARR is 7% per year. Determine the Annual Worth (cost) of this de-icing rig, and briefly explain why the Do-Nothing alternative is not an option in this situation.

Exercise 3: Comparing Equal-Life Alternatives: AW and PW Together

A university is deciding between two mutually exclusive options for a new campus solar array, each with a useful life of 6 years. The university's MARR is 8% per year.

- **Option 1:** Initial cost \$720,000, salvage value \$60,000, generates annual net savings (on electricity bills) of \$185,000.
- **Option 2:** Initial cost \$480,000, salvage value \$30,000, generates annual net savings of \$118,000.

1. Compute the Present Worth (PW) of each option over its 6-year life.
2. Convert each PW value directly into an Annual Worth (AW) using the appropriate $(A/P, i, n)$ factor.
3. Now compute the AW of each option independently, using the Capital Recovery formula directly (i.e., without first computing PW). Confirm that your two approaches to finding AW agree.
4. Based on your results, which option should the university select, and why?

Exercise 4: Independent Projects Screening via Annual Worth

A food-processing plant has identified three independent efficiency projects, each with a useful life of 6 years and no budget constraint this year. The plant's MARR is 10% per year.

- **Project 1:** Initial cost \$85,000; generates annual net savings of \$21,000.
- **Project 2:** Initial cost \$140,000; generates annual net savings of \$29,000.
- **Project 3:** Initial cost \$45,000; generates annual net savings of \$11,500.

1. Compute the Capital Recovery (CR) and Annual Worth (AW) of each project.
2. State which project(s) the plant should implement.

Exercise 5: Mutually Exclusive Cost Alternatives with Unequal Lives

A city's public works department must select one of two wastewater pumps to keep a lift station operating; the service is mandatory, and one of the two pumps must be chosen. Pump A costs \$260,000, has a useful life of 5 years, no salvage value, and an annual operating cost of \$38,000. Pump B costs \$410,000, has a useful life of 8 years, and a salvage value of \$25,000; because Pump B is a larger, more complex unit, its maintenance

burden grows as it ages: its operating cost starts at $15,000 in Year 1 and increases by 8% each year through Year 8. The department's MARR is 6% per year. Using Annual Worth analysis, determine which pump the department should select. You do not need to extend the cash flows to the Lowest Common Multiple of the two lives.

Exercise 6: Sensitivity of Capital Recovery to Salvage Value

A logistics company is purchasing a fleet delivery van for $65,000, with a useful life of 6 years and a MARR of 8% per year. The company's finance team is uncertain about the van's resale value at the end of its life, and asks you to evaluate the Capital Recovery (CR) under two different salvage value estimates: a conservative estimate of $8,000, and an optimistic estimate of $14,000.

1. Compute the CR under the conservative salvage estimate.
2. Compute the CR under the optimistic salvage estimate.
3. Determine the percentage difference between the two CR values, and briefly comment on how sensitive Capital Recovery is to changes in the salvage value estimate in this case.

Exercise 7: Annual Worth of a Permanent Investment

A coastal municipality is planning a permanent seawall to protect its waterfront, with an indefinite useful life. The MARR used for public infrastructure projects is 5% per year. The estimated costs are:

- Initial construction cost at Year 0: $6,500,000
- Routine annual inspection and upkeep: $85,000 per year, continuing indefinitely
- A non-recurring reinforcement of the wall's foundation at Year 8: $400,000
- A recurring resurfacing of the wall's protective coating every 15 years, at a cost of $350,000 each time (first resurfacing at Year 15)

1. Determine the total equivalent Annual Worth (cost) of the seawall in perpetuity.
2. Explain, in plain terms, what this total annual figure means for the municipality's long-term budgeting.

Exercise 8: Relating Capitalized Cost to Annual Worth

A university is comparing two ways of funding a permanent scholarship endowment with an indefinite life, at a MARR of 6% per year. Financial Option 1 is described in terms

of its Capitalized Cost: it requires a Capitalized Cost of $2,400,000. Financial Option 2 is described in terms of its Annual Worth: it requires an equivalent annual contribution of $135,000 in perpetuity.

Using the relationship between Capitalized Cost and Annual Worth for a permanent investment, determine which of the two funding options requires a larger equivalent commitment from the university, expressing both options in the same terms (either both as Capitalized Cost or both as Annual Worth) before comparing them.

Exercise 9: Verifying Single-Life-Cycle AW Against LCM Present Worth

A mining company is comparing two mutually exclusive haul truck models. Truck X costs $310,000, has a useful life of 5 years, a salvage value of $35,000, and generates annual net savings of $98,000. Truck Y costs $450,000, has a useful life of 8 years, a salvage value of $50,000, and generates annual net savings of $112,000. The company's MARR is 9% per year.

1. Compute the Annual Worth (AW) of each truck model.
2. Separately, determine the PW of each alternative; use LCM approach if necessary.
3. Convert each of your PW values from part 2 into an Annual Worth. Confirm that these values match your AW values from part 1.
4. Based on your results, which truck model should the company purchase?

Exercise 10: Integrated Capstone: CapEx, OpEx, and a One-Time Cost

A hospital is evaluating a new sterile processing system for its surgical department. The system requires an initial investment (CapEx) of $920,000, has a useful life of 7 years, and a salvage value of $60,000 at the end of its life. The system is expected to generate annual net savings (OpEx reduction) of $210,000 per year. In addition, a mandatory mid-life recalibration and compliance inspection is required at the end of Year 4, at a one-time cost of $45,000. The hospital's MARR is 8% per year.

1. Determine the equivalent annual cost contribution of the one-time Year 4 recalibration cost, converting it into an annual amount over the system's full 7-year life.
2. Compute the Capital Recovery (CR) of the system, accounting for the initial investment and salvage value. Is the recalibration cost part of the CR?
3. Compute the total Annual Worth (AW) of the system, combining the Capital Recovery, the annual net savings, and the equivalent annual cost of the Year 4 recalibration.
4. Using the AW value you found in part 3, determine the equivalent Present Worth (PW)

of the entire system over its 7-year life, and confirm that this PW is consistent with directly discounting all of the system's original cash flows (CapEx, salvage, annual savings, and the one-time Year 4 cost) to the present.

5. Based on your results, should the hospital proceed with the sterile processing system?